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The Economics of the Global Energy Challenge†

By Michael Greenstone*

Rather than facing an isolated climate change challenge, this paper argues that the world must confront the Global Energy Challenge (GEC) that requires all countries to make trade-offs between three often competing and interrelated goals: inexpensive and reliable energy, clean air, and limiting damages from climate change. This paper presents seven facts that help illuminate the contours of the GEC and the interactions between the three goals. It concludes by outlining potential solutions: pricing energy at its full social cost, investing in technical and policy innovation, improving information on pollution and climate damages, and treating energy as a private good.

Climate change is having a moment, and not just because my kids talk about it almost as much as they talk about Taylor Swift. The year 2023 was the hottest ever recorded, and temperatures are expected to continue to rise (Cohen 2023). The emerging field of climate attribution is pinpointing the precise role that greenhouse gas (GHG) accumulation plays in individual storms, heat waves, flooding, etc. (IPCC 2023). The United Nations Climate Change Conference was just hosted by the United Arab Emirates, a major oil and gas exporter, and yet still concluded that countries need to "transition away" from fossil fuels. If that weren't enough, not only did the United States recently quadruple its estimate of the social cost of carbon (SCC)—a measure of climate damages that helps determine the stringency of regulations to restrict GHG emissions (EPA 2023)—the 2022 Inflation Reduction Act marked our country's boldest climate legislation to date.[1]

My thesis is that we would be better off if, instead of just climate change, the global energy challenge were having a moment to compete with Taylor Swift for our attention. The global energy challenge is defined by three often conflicting goals that all societies are pursuing: inexpensive and reliable energy, clean air, and limiting damages from climate change. Although we often think of energy as a homogeneous good, there are many energy sources (e.g., coal, natural gas, nuclear, solar, wind, and petroleum) that differ in their price per unit of power, effects on conventional or local air pollution,[2] and effects on the accumulation of GHGs that cause climate change in the atmosphere.[3] In most settings, no

* Department of Economics, University of Chicago; National Bureau of Economic Research; Centre for Economic Policy Research (email: mgreenst@uchicago.edu). I thank Marianne Bertrand, Amy Finkelstein, Rema Hanna, Trevor Houser, Bob Kopp, Kate Larsen, Ben Olken, Rohini Pande, Nick Ryan, and Anant Sudarshan for insightful comments that greatly improved the lecture and this paper. Outstanding research assistance was provided by Kate Champion, David Du, Jonah Gilbert, Neil Himwich, Aliya Jabbar, Matthew Neils, Noah Sobel-Lewin, and Sariya Stowers. I am additionally grateful to the Rhodium Group for sharing data from their *Rhodium Climate Outlook* (Larsen et al. 2023).

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[1] Of course, it bears noting that United States' climate exceptionalism remains intact as we are the only G7 country without a national carbon pricing policy.

[2] The combustion of fossil fuels to produce energy is a primary source of conventional or local air pollution. The air pollutants of especial concern for public health are particulate matter (PM), carbon monoxide, ozone, nitrogen dioxide, and sulfur dioxide, with PM widely considered the most pernicious. As a general matter, conventional or local air pollution is treated as a within-country problem. However, Heo, Ito, and Kotamarthi (2023) demonstrate that local air pollution and its health effects do not perfectly respect international borders.

[3] The combustion of fossil fuels to produce energy is also a primary source of GHGs (e.g., CO_2, methane, and nitrous oxide). GHGs' primary risk comes from increasing

single energy source is the cheapest, the smallest contributor to conventional air pollution, and the least impactful on climate change. For example, although fossil fuels (e.g., coal, oil, and natural gas) are a primary contributor to conventional air pollution *and* GHGs, different fossil fuels have different costs and differ in the rate at which these air pollutants and GHGs are emitted per unit of energy. Thus, choosing an energy source involves trading off across these three important determinants of well-being. Focusing on only one of those characteristics (e.g., climate change) misses the full impact on human well-being or welfare.

Trade-offs are always difficult. However, those involved in the global energy challenge are especially so because GHGs have the same impact on climate change wherever they are emitted (e.g., Norway, China, or Pakistan). Yet different countries will make different choices because the trade-offs between the three goals vary across and within countries based on wealth, geography, current and projected future climate, values, functioning of the political system, etc. Therefore, decisions made in Norway may not be ideal for China or Pakistan and vice versa.

The global energy challenge's requirement that all countries balance these three goals, with the goal dealing with climate impact involving complete interdependence, makes it perhaps the defining challenge of this century. I now describe each of these three goals in some more detail.

The first goal is access to inexpensive and reliable energy. Energy does not provide any direct benefits, but rather demand for it is derived from the goods (e.g., steel bars) and services (heating and cooling) that require energy for their production. This means that every $1 spent on energy embedded in the products that we consume reduces our consumption of other goods, like food, shelter, healthcare, and so forth, that directly enter the utility function by $1. Furthermore, higher energy prices make production more expensive, and this can reduce wages and purchasing power (Brown and Yücel 2002). Moreover, reliable energy is essential for everything from advanced manufacturing to hospital services to temperature control in the face of extreme weather. In short, energy's cost and reliability govern the quality of a person's life. It is therefore not surprising that there are not any examples of countries becoming rich without consuming high levels of energy.

The second goal of the global energy challenge is to limit the damages from local air pollution due to energy consumption. It may be surprising, but air pollution is the greatest current external threat to human health globally, with the average person losing more than two years of life expectancy from air pollution. This loss is comparable to that from tobacco smoking and much greater than that from alcoholism, terrorism, war, and so on (Greenstone and Hasenkopf 2023). The air-pollution-induced loss of life expectancy varies widely around the globe, with relatively low levels in today's wealthy countries and losses of four years or more in India, Bangladesh, Pakistan, and other parts of South Asia (Greenstone and Hasenkopf 2023). Energy production that limits local air pollution is typically more expensive largely because coal is bountiful and inexpensive compared to combining coal combustion with air pollution control devices[4] or using alternative energy sources. In the context of the global energy challenge, it is noteworthy that the costs and benefits of policies that address the air pollution externality largely occur within the same country.

The nature of the costs and benefits of reducing GHG emissions makes it difficult to achieve the global energy challenge's third goal of limiting climate change damages. On the cost side, the willingness to pay (WTP) for low GHG energy sources varies greatly across countries and is much lower in today's poor and rapidly growing countries and regions (e.g., India and sub-Saharan Africa) in part because they will be richer in the future (meaning they have a relatively high marginal utility of consumption

the amount of heat held on the earth's surface, which causes climate change. Additionally, they are considered global pollutants because their impact on warming is unaffected by where they are emitted, while the effects of conventional air pollutants are much more local.

[4] Air pollution control devices are used widely to reduce emissions of local air pollutants. In countries with stricter air pollution laws and regulations, more effective (and expensive) technologies are frequently required for plants that use coal and other fossil fuels to produce energy. In contrast, the technologies to capture GHGs (e.g., carbon capture and sequestration (CCS)) before they enter the atmosphere are generally prohibitively costly, less mature, and used infrequently around the world.

today).[5] One then must add to this that today's poor and rapidly growing countries are projected to account for the vast majority of future emissions. The cruelty of climate arithmetic is quickly apparent—for example, limiting temperatures increases to the 2.0° C increase recommended by scientists requires non-OECD (Organisation for Economic Co-operation and Development) countries to cut their projected end-of-century CO_2 emissions by roughly 85 percent. Put more plainly, the prospects for limiting climate change are greatly diminished by the fact that reducing global emissions requires the largest emissions cuts to occur in the countries where the WTP for them is lowest.

Features of the benefits of cutting GHG emissions also make limiting climate change difficult. One is that the benefits of such reductions are highly heterogeneous (Carleton and Greenstone 2022; Nath et al. 2024). Reductions in emissions are projected to be especially *beneficial* in areas that are already hot or relatively poor and especially in those that are both hot and poor (e.g., Pakistan and Bangladesh). In contrast, emissions cuts might naturally be low priority in places where higher temperatures would be beneficial (e.g., Northern Canada, Siberia, and Alaska). It is at least plausible, and perhaps even likely, that countries' WTP for reducing GHG emissions today will be based on projected damages in their country, rather than average global damages, adding a further layer of complexity to efforts to limit damages (Golosov et al. 2014). Beyond the heterogeneity in impacts across space, the benefits of cutting emissions emerge over centuries, and this raises thorny questions of countries' discount rates over long time horizons.

An efficient effort to limit climate change therefore requires countries with heterogeneous costs and benefits of cutting GHG emissions to execute a common plan to cut GHG emissions.[6] In a frictionless world, a benevolent social planner could undoubtedly create a global market that would restrict GHG emissions efficiently and would undoubtedly include large transfers. However, that world only exists in Econ 101 textbooks and university classrooms. In reality, there is no global social planner, benevolent or otherwise, and the world is characterized by frictions, political economy challenges, and countries pursuing their own interests. So, although GHG emissions have the same effect on the climate no matter where they take place (e.g., Oslo, Beijing, or Lahore), decisions about GHG emissions reflect local conditions, not global efficiency. Or to borrow from Tip O'Neil, former speaker of the US House of Representatives, who said "All politics are local," all decisions about GHG emissions are local.

The world is grappling with a global energy challenge that requires all countries to balance the needs for inexpensive and reliable energy, clean air to breathe, and managing climate risk when choosing their energy sources. There is no single right way to handle this balancing act because the trade-offs between these often conflicting goals differ widely across countries and even within them. The resulting differences in choices would not merit such careful consideration except that GHGs are a global pollutant that does not respect borders, making every country dependent on every other.

The remainder of the paper is divided into three sections. The next section of the paper illustrates the nature of the global energy challenge. Specifically, I present seven facts about the trade-offs inherent in the quest to balance these three goals.

Section II identifies four market failures that plague energy markets and discusses solutions to them. The market failures are that energy prices generally fail to reflect social costs (e.g., due to air pollution and GHG emissions); innovation, both technical and policy, is a public good and consequently undersupplied; imperfect information prevents people from optimizing in response to environmental quality; and energy access is compromised by the treatment of electricity as a right in many developing countries.

Finally, Section III summarizes the paper and concludes.

[5] WTP for GHG reductions fundamentally involves choosing more expensive energy today, which reduces current consumption in exchange for smaller climate damages in the future. Thus, WTP is determined by the ratio of marginal WTP for consumption today and in the future (Becker, Murphy, and Topel 2010). This ratio varies greatly across countries—the relative value of a dollar of consumption today in today's poor and rapidly growing countries is much higher than in rich countries, precisely because they are poor relative to rich countries today and relative to their futures.

[6] All of this is part of the reason that the great economist Martin Weitzman referred to climate change as "the problem from hell" (Weitzman 2014).

I. The Global Energy Challenge and Its Consequences

Section A presents seven key facts that illustrate the global energy challenge. Section B then discusses the implications of these facts and their meaning for human well-being, mediated through energy prices, exposure to air pollution, and risks from climate change.

A. *Seven Facts That Define the Global Energy Challenge*

Fact 1: Energy Consumption Is Critical for Living Standards

The first fact is that energy is critical for improving living standards and well-being. Figure 1, panel A plots log GDP per capita (vertical axis) against log total energy consumption per capita (horizontal axis). The life of an empirical economist is filled with data that disappoint by failing to provide a precise answer or contradicting a closely held hypothesis. But this figure is a rare exception because it has such a clear and powerful message. Simply put, the world does not offer any examples of a country achieving high income levels without consuming lots of energy.

Why is this the case? At its core, energy is a substitute for using the human body to produce output, transport, comfort, and so forth. Machines powered by energy have freed us from backbreaking work in the fields, dangerous and taxing manufacturing work, and so much more. The heating and cooling that make us comfortable and protect us from dangerous temperatures also require energy (Barreca et al. 2016). Further, the transportation that allows us to go to faraway places or to consume goods produced in those places requires energy too.

Per capita income can be an antiseptic way to describe differences in living standards, however, so Figure 1, panel B provides an alternative. It is taken from Burgess et al. (2024), which examines the impacts of hot days on mortality. It's primarily focused on India, but there's also a comparison with the United States for reference. This figure reports the impact on annual mortality rates of a day at different average temperature ranges (which equal the mean of the maximum and minimum) relative to a day that's between 70° and 74° F.

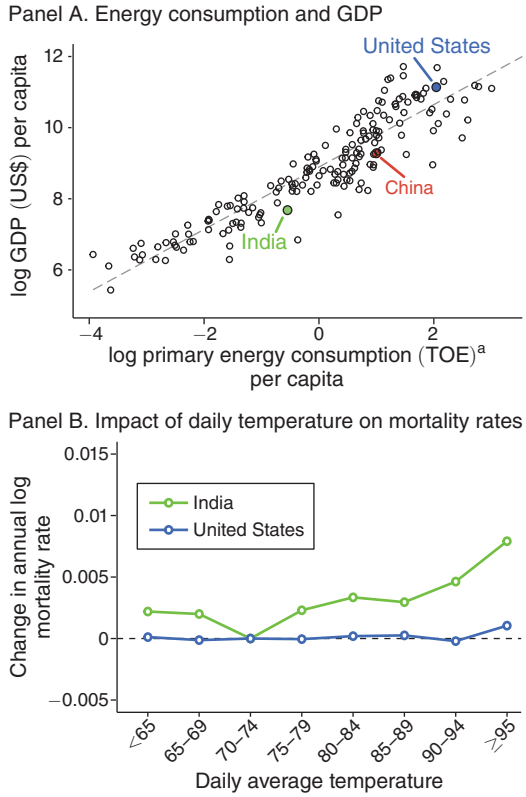


Figure 1. The Relationship between Energy and Growth

Notes: Individual points in panel A reflect country-specific GDP-energy consumption values. Data on energy consumption come from EIA (2024). Data on GDP and population come from World Bank (2024a, c). The gray dotted line is the linear best fit through the data. Panel B reflects mortality estimates from Burgess et al. (2024). Plotted lines report seven coefficient estimates (circle markers) of the effect of an additional day in the corresponding temperature bin relative to the effect of a day in the 70–74°F reference bin on the annual all-age log mortality rate. The India regression controls for rainfall (upper/lower tercile indicators), district fixed effects and year fixed effects, and separately for rural and urban areas. The US regression controls for rainfall (upper/lower tercile indicators), state-by-month fixed effects and year-by-month fixed effects. The regressions are weighted by the district population (India) and state population (United States) in 1991. The mortality data for the United States are taken from Barreca et al. (2016) as well as the CDC. Standard errors are clustered on the district (India) and the state (US) level.

[a]The unit tons of oil equivalent (TOE) represents the amount of energy released by burning one ton of crude oil. Per capita mean consumption was 2.03 TOE globally in 2022. In the same year, US energy consumption per capita was 7.48 TOE.

What happens on hot days? The answer depends on where you live, with the green line plotting India's mortality response function and the blue one the United States'. When a hot day arrives in India, it is truly devastating; for example, a single day that is above 95° F raises the *annual* mortality rate by 0.8 percent. In contrast, there is virtually no relationship between daily temperatures and annual mortality rates in the United States when the US response function is viewed on a scale that accommodates the India one. In practice, the estimated impact of a day above 95° F is about an order of magnitude more deadly than a day with the same temperature in the United States.

If you're in India and you're on the green line and you see the United States on the blue line, then you know a fuller and safer life is possible. The green line reflects numbers and numbers can be cold and scientific, but these numbers represent sickness, shorter lives, and, ultimately, lives that are compromised in countless ways by poverty. The best way for India to get to the blue line is almost certainly by growing incomes, and that requires a lot more energy consumption. This is because higher incomes unlock a multitude of ways to protect against temperature extremes, certainly including air conditioning (Barreca et al. 2016).

Overall, my interpretation of these data is that any solution to managing the trade-offs inherent in the global energy challenge must accommodate continued global growth in energy per capita. For example, any efforts to convince societies with low levels of energy consumption that they are already consuming enough seems certain to fall on deaf ears, because they are rightly trying to ensure their citizens can have better, healthier, and more productive lives. Indeed, some interpret rich countries' efforts to push developing countries to restrict their GHG emissions as "climate imperialism" (Subramanian 2022). Similarly, today's wealthy countries are reluctant to reduce their energy consumption in a meaningful way as they too want to advance living standards.

Fact 2: Energy Consumption Is Unequal

Fact two is that there is substantial inequality in energy consumption. Figure 2 shows per capita electricity consumption in the Indian state of Bihar, all of India, China, and the United States.

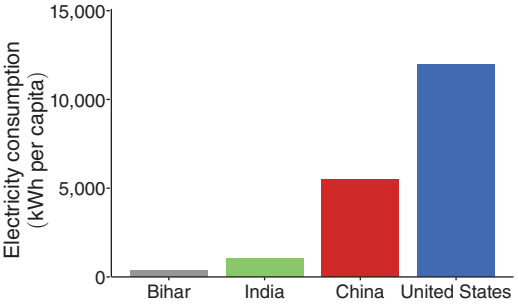


Figure 2. Per Capita Electricity Consumption

Notes: Data on electricity consumption for India, China, and the United States were obtained from EIA (2022). Data for Bihar come from the Government of Bihar Finance Department (2021–2022).

In the United States, the average person uses about 13,000 kilowatt hour (kWh) of electricity per year, and there are about 330 million people in the United States. In China, that number is approximately 5,000 kWh of electricity per year, and its population is 1.4 billion people. The annual per capita electricity consumption in India, also with a population of 1.4 billion, is roughly 1,000 kWh.

I include the Indian state of Bihar separately because it is such a telling example. Its population is roughly 131 million people (about 40 percent of US population) and its per capita electricity consumption was 200 kWh in 2021. For context, it takes 131 kWh to use a 60-watt (conventional) light bulb for 6 hours per day for a full year. This is one of those numbers that reminds me of my late and great colleague Robert Lucas's famous line that "Once you start thinking about [economic] growth, it's hard to think about anything else." The point is that 200 kWh per year represents lives where manual labor does much work that we in rich countries assume can only be done by machines, leading to injuries, near complete exposure to weather extremes, and so forth.

How pervasive is "low" energy consumption? 3.85 billion people live in countries with per capita electricity consumption below 1,500 kWh per capita annually. And 4.36 and 6.83 billion people live in countries with consumption below 2,500 kWh and 5,500 kWh per capita annually, respectively. The point is that there are billions of people who want the higher energy consumption that helps to unlock higher living

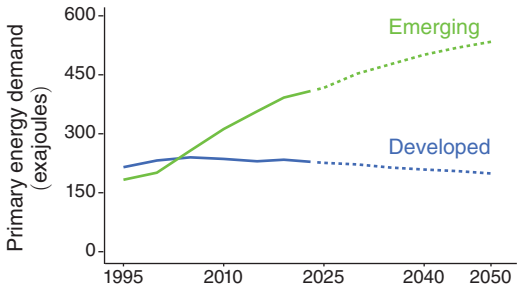


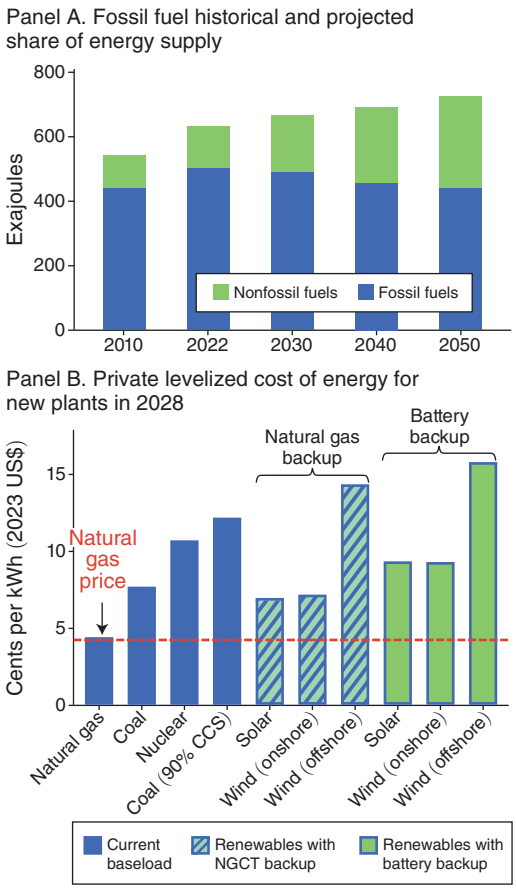
Figure 3. Historical and Projected Energy Demand in Emerging and Developed Countries

Notes: Emerging and developed demand projections were obtained from Dale (2023, 13). Projections use the "New Momentum" scenario, which BP describes as "designed to capture the broad trajectory along which the global energy system is currently travelling. It places weight on the marked increase in global ambition for decarbonization in recent years, as well as on the manner and speed of decarbonization seen over the recent past. CO_2e emissions in New Momentum peak in the 2020s and by 2050 are around 30 percent below 2019 levels."

standards and to consume the products with higher embodied energy intensity that people in rich countries consume.

Fact 3: Energy Consumption Is Projected to Grow Rapidly in Emerging Economies

The third fact, which almost naturally follows from the first two, is that global energy consumption is expected to grow by 14 percent between 2025 and 2050 (Dale 2023). Figure 3 presents historical and projected primary energy demand, separately for the "developed" countries (2023 population of 1.2 billion) and the rest of the world (2023 population of 6.8 billion).[7] Among "emerging" countries, energy consumption is projected to grow by 28 percent from 2025–2050 compared to a 12 percent decline among "developed" countries due to greater efficiencies. In total, the "emerging" countries account for 130 percent of the expected growth. More granularly, India's primary energy growth (45 exajoules,[8] 108 percent) is projected to account for over half of global growth in energy demand (88 exajoules), and Africa's energy demand is projected to grow by 62 percent between 2025 and 2050.



(*continued*)

Figure 4. Fossil Fuel Use Will Continue to Increase

Fact 4: Fossil Fuels Are Expected to Remain the Dominant Source of Energy

The news is filled with stories about the declines in the costs of renewable electricity sources. These declines are indeed newsworthy since they have been large and unexpected. At the same time, the declines in the costs of recovering petroleum and natural gas have also been substantial. Fact 4 is that fossil fuels are projected to remain the dominant source of energy, at least through the middle of the century.

Overall Energy Consumption.—Figure 4, panel A illustrates that the International Energy

[7] BP defines developed countries as those in North America, Europe, and OECD Asia.

[8] For reference, global consumption of energy was 634 exojoules in 2022.

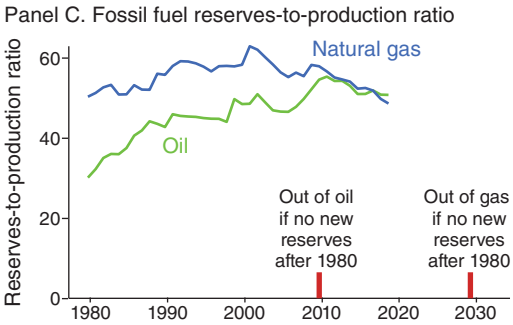


FIGURE 4. FOSSIL FUEL USE WILL CONTINUE TO INCREASE (*continued*)

Notes: Panel A shows historic and projected total energy supply by fossil status of source from IEA (2023b). Fossil fuels are gas and oil. Panel B shows the LCOE of fossil fuels and nuclear (blue bars), renewables with battery backup, and renewables with natural gas combustion turbines (NGCTs) as backup (green bars with blue hash). NGCTs offer an inexpensive way to flexibly generate energy with low lead times. Coal (90 percent CCS) is a coal plant with CCS technology capable of capturing 90 percent of carbon emissions. Further details of the calculation of LCOEs are in the notes of Table 2. Panel C uses reserve-to-production ratios from (2021), the last year in which reserve-to-production ratios for Natural Gas were published. The "Out of oil/gas if no new reserves after 1980" segments are drawn at the years when, assuming constant production/demand at 1980 levels, the 1980 reserves would have been fully depleted.

Agency estimates that today about 79 percent of all energy comes from fossil fuels. In 2050, they still project it to be 61 percent (IEA 2023b). Why? Because, as detailed directly below, fossil fuels are remarkably inexpensive when measured by the private costs of production.

Power Sector.—Figure 4, panel B reports the levelized cost of electricity (LCOE) in cents per kWh associated with different types of new power plants that come online in 2028.[9] The levelized cost is the price that a plant must charge such that the sum of present-discounted annual revenue from generation is equal to the sum of present-discounted annual costs;[10] this is a standard approach to determining a plant or technology's economic viability. Finally, I note that these estimates are averages for the United States and will vary across and within countries based on local cost factors (e.g., availability of inexpensive natural gas).

The technologies in Figure 4, panel B are divided into three groups and denoted by color. Those in blue are considered "baseload" technologies because they operate continuously except for maintenance periods. In practice, coal, combined cycle natural gas, and nuclear plants have average operation rates of 85 percent, 87 percent, and 90 percent, respectively.

Among the baseload technologies, natural gas is easily the least expensive. It costs just 4.3 cents per kWh and is the standard against which investment decisions about other sources are made.[11] At 7.6 cents per kWh and 10.6 cents per kWh, respectively, coal and nuclear plants are about 75 percent and 150 percent more expensive than natural gas.[12] It is clear from the per unit economics that investments in these technologies are unattractive on average, given that electricity markets are structured to choose sources predominantly on private costs. The figure also reports private costs of 12.1 cents per kWh for a new coal plant equipped with carbon capture and sequestration (CCS) technology that captures 90 percent of a coal plant's typical CO_2 emissions. Without substantial subsidies, this type of plant would be far out of the money.

The next two sets of bars consider the LCOEs of zero-carbon renewable energy sources when they are paired with backups so that they can be properly compared to the baseload technologies on an apples-to-apples basis. The popular press frequently talks about these technologies' costs but ignores the costs of the additional backups

[9] The year 2028 was chosen because it takes about four to five years to complete a nuclear plant, and the aim was to make an apples-to-apples comparison.

[10] They are calculated by summing present-discounted annual costs including construction costs, operation and maintenance costs, fuel costs, and transmission costs. For nuclear, waste disposal and decommissioning costs are included as well. Annual generation is determined by the plant's capacity and the technology's average operation rate.

[11] It is noteworthy that the LCOE for a new natural gas plant in 2005 was 8.5 cents (also in 2023 US dollars) (EIA 2006). The decline is largely due to the hydraulic fracturing revolution that dramatically reduced the costs of recovering natural gas, underscoring the continuing innovation in fossil fuel recovery.

[12] The coal LCOE of 7.6 cents per kWh is for a coal plant that is required to install expensive pollution abatement equipment to prevent the release of sources of local air pollution as per United States Clean Air Act regulations. The LCOE for a coal plant without this abatement equipment, like the plants in many lower-income countries, would be substantially lower.

that using them requires to fill in for the periods when the sun is not shining or the wind is not blowing. Indeed, these technologies are quite intermittent, as utility-scale solar power plants in the United States operate at about 29 percent of their electricity-generating capacity and onshore wind at roughly 40 percent. It is apparent that these plants need substantial back-ups to be compared to the baseload technologies (EIA 2023). We use the effective load carrying capability (ELCC), an industry standard that captures the degree to which a new source contributes to overall reliable grid capacity, to quantify the amount of backup generation/storage that must come online simultaneously with a renewable source (see the online Appendix for further details).

The middle set of bars (colored green with cross hatches) assumes that the renewables' down periods are filled in with a natural gas peaker plant, which can ramp up and down quickly and inexpensively. These plants are commonly used to address supply shortfalls and demand peaks in the United States. This approach of pairing renewables with gas peaker plants yields LCOEs of 6.8, 7.0, and 14.3 cents per kWh for solar, onshore wind, and offshore wind, respectively. Both solar and onshore wind are competitive with coal but meaningfully more expensive than a baseload natural gas plant. It is noteworthy that these pairings of renewables with gas peaker plants are not carbon free since the peaker plants run such a high fraction of the time.

The third set of bars (colored green) assume that the renewables are paired with utility scale batteries to smooth out the intermittencies. Despite the substantial progress in reducing battery costs (IEA 2023a), this is an expensive way to produce electricity with LCOEs of 9.2, 9.2, and 15.7 for onshore wind, solar, and offshore wind, respectively. These LCOEs are more than double the cost of a baseload natural gas plant. However, these sources are zero carbon technologies.

Figure 4, panel B therefore powerfully illustrates that, in the power sector, fossil fuels are the least expensive source of electricity when judged by the private costs of production. Since we largely structure energy markets to choose the source with the lowest private cost, it is not surprising that fossil fuels are projected to remain dominant. It is worth underscoring, though, that the structure of energy markets is not a law of physics or a religious edict—it is a decision about the world that we want to have. It is therefore disingenuous to choose the structure and then be surprised by the consequences. Section IIA explores how pricing energy at its full social costs, so that the air pollution and GHG emissions externalities are internalized, would change Figure 4, panel B's conclusions.

Transportation Sector.—We also use energy in the transportation sector, and here too fossil fuels are less expensive than the alternatives when judged on private costs. Based on an approach from Covert, Greenstone, and Knittel (2016), I compared the lifetime ownership costs of otherwise comparable electric vehicles and internal combustion engine vehicles that run on petroleum. I found that, for lifetime ownership costs to be equal, oil prices would have to be $129, about 67 percent higher than their current price of $77. Or alternatively, the price of batteries would have to come down by 35 percent for EVs to be cost competitive with ICE cars at today's oil prices. So, while there has been a lot of progress with EVs, they are still more costly, even without accounting for the costs associated with range anxiety, charging availability, and battery disposal.

Will We Run Out of Fossil Fuels?—But will fossil fuels also be cheap in the future? It is true that there are a finite amount of fossil fuels on the planet, and that has caused some to think that, while they are cheap today, they will be in short supply and expensive in the future (Hubbert 1956).[13] What this style of thinking has missed is that there has been tremendous innovation in reducing the costs of recovering fossil fuels, and there is little reason to think innovation will stop going forward. The fracking revolution was a step change for oil and gas, but there is also methodical year-by-year grind-it-out innovation that reduces costs (Covert 2015).

Figure 4, panel C captures the consequences of this innovation by plotting the

[13] One famous example comes from Marion King Hubbert's prediction in 1956 that global oil production would peak at 12.5 billion barrels per year around 2000 (Hubbert 1956). This prediction proved to be far from accurate as production was 27.2 billion barrels in 2000 and continues to rise, reaching 34.2 billion barrels in 2022.

reserves-to-production ratio for oil and natural gas. Reserves sounds like a physical concept, but it is actually closer to an economic one. Specifically, it requires that a mineral resource "can be recovered in the future from known reservoirs under existing economic and operating conditions" (BP 2021). Thus, the reserves-to-production ratio provides a measure of how many years current reserves will last at current rates of production (note that production and consumption are tightly linked because storage is very expensive). So, for example, there were about 30 years of oil reserves available in 1980 with 1980 production levels. Thus, a static view of reserves in 1980 would have indicated that the world would run out of oil by 2010 (or earlier since production grew by 32 percent between 1980 and 2010 (IEA 2021)). However, this ratio was even higher at 53 years in 2020! It is similarly noteworthy that the natural gas ratio has held constant, even as production increased by 115 percent between 1980 and 2020 (IEA 2021). Additionally, the third main fossil fuel, coal, is not shown here, but its ratio is 139 years in 2020, making clear that there will not be a shortage of coal over any climate-relevant timeline.

It is apparent that the engineers, geologists, and so forth at the fossil fuel companies are not sitting on their hands and indeed are remarkably innovative. I rate it as extraordinarily likely that inexpensive fossil fuels will remain widely available far into the future. Further, the history of people leaving what amounts to $100 bills in the ground unperturbed is very short, bordering on nonexistent. Thus, the future of fossil fuels is likely to remain bright as long as energy markets are structured to choose the source with the lowest private costs.

Fact 5: Fossil Fuels Increase Air Pollution That Shortens Lives

So far, the facts that I have presented focus on the first goal of access to inexpensive and reliable energy. Now I turn to the second goal: managing the damages of conventional or local air pollution. These air pollutants are not the ones generally causing climate change, but rather they affect human health, aesthetics, and a range of other determinants of well-being. Indeed, they were a primary motivation for the 1970 establishment of the Environmental Protection Agency (EPA) in the United States and the target of the landmark 1970 Clean Air Act.

Fact 5 is that the combustion of fossil fuels increases conventional air pollution, especially airborne particulate matter (PM) (McDuffie et al. 2021). About 7.6 billion people around the world live in areas with PM concentrations that the World Health Organization (WHO) considers unsafe.[14]

What do we know about the effects of long-run exposure to air pollution? There is a large literature on the effects of short-term exposure to air pollution (e.g., Holub and Thies 2023), but the long-run consequences are more important and harder to answer (Dominici, Greenstone, and Sunstein 2014). The gold standard approach to researching this question would be to randomly assign one set of people to different levels of air pollution over the course of their entire lifetimes. It would then be possible to observe the health consequences, along with the compensatory behaviors and expenditures that people undertake to protect themselves (e.g., Deschênes, Greenstone, and Shapiro 2017). This would allow us to estimate a full measure of the welfare consequences of high levels of air pollution. Thankfully, no university institutional review board, much less a government, would allow such a study to take place.

An appealing alternative, though, is to find a natural experiment (Nobel Prize in Economic Sciences 2021) that mimics such a true randomized control trial (RCT). In 2013 and 2017 papers, some collaborators and I identified China's Huai River winter heating policy as a powerful natural experiment that could potentially shed light on the long-run consequences of air pollution exposure (Chen et al. 2013; Ebenstein et al. 2017). The winter heating policy was established during the planning period when China was much less wealthy than today and didn't have the resources to heat every home during the winter. Specifically, the policy drew a line across the middle of the country that

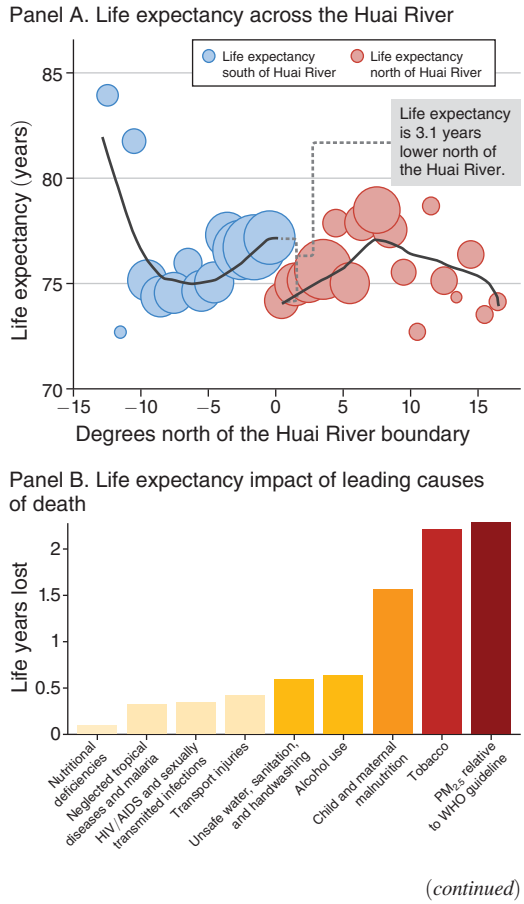
[14] This calculation was done using the 2021 $PM_{2.5}$ and population data available at aqli.epic.uchicago.edu. Specifically, 7.6 billion people live in level 2 global administrative areas (e.g., a United States county) where the average annual $PM_{2.5}$ concentration exceeds the WHO standard of 5 micrograms/m, and this accounts for about 98 percent of the global population.

follows the Huai River and Qinling mountain range: areas to the north of the line received free winter heating. The heating was supplied by small coal boilers without pollution controls that were attached to most buildings in the North (i.e., these were not efficient power plants with pollution controls situated away from population centers). Further, people were provided free coal from November 15 to March 15 every year. Areas to the south were out of luck and denied winter heating.

This policy went on for several decades and indeed its vestiges remain today (although market reforms and a shift to natural gas in the last few years are changing it). The policy thus lends itself to a regression discontinuity design that compares people born just to the north of the river to people born just to the south. Importantly, the papers find that PM concentrations are discretely 40 percent higher on the north side of the river, demonstrating that the policy was adhered to. It has many appealing features as a natural experiment, which are detailed in the papers.[15]

Figure 5, panel A reproduces the key finding that life expectancy is about three years lower just to the north of the line relative to just to the south. Each of the bubbles are 1° latitude displacements from the Huai River and are proportional in size to the region's population. So the first red (blue) bubble to the right (left) of the vertical line at 0 represents people who live between 0° and 1° north (south) of the river. The jump downward in life expectancy is not evident at any other degree displacements. Further, it all comes from cardiorespiratory causes of death that are plausibly related to air pollution, as there are not discrete differences in the incidence of other causes of death at the river's edge. Overall, the result is that this quasi-experiment provided a rare opportunity to estimate the long-run consequences of exposure to PM air pollution.

Based on this research, I created the Air Quality Life Index (AQLI) (Greenstone and Hasenkopf 2023) that takes the elasticity of life expectancy with respect to PM from the Huai paper and combines it with highly geographically resolved satellite data on PM concentrations. The AQLI combines them to provide information on the life expectancy consequences of moving from current concentration to a country's air pollution standard or the WHO standard for every community globally. It assumes that the relationship between life expectancy and PM is linear.[16]



(*continued*)

FIGURE 5. AIR POLLUTION CAUSES SIGNIFICANT AND HETEROGENEOUS LOSS OF LIFE

[15] For example, until the last 10–15 years, migration was highly restricted due to the Hukou system, so if people were observed in the north or south, one could have a high degree of confidence that they had spent their life there. Further, the line was not used for any other policy and does not reflect jurisdiction (e.g., provinces) boundaries.

[16] Approximately 1.9 billion people live in a location that is between the 5 and 95 percentiles of the $PM_{2.5}$ distribution in the Huai River paper's sample, suggesting that there is empirical support for applying results from Ebenstein et al. (2017), especially at relatively high concentrations. At the same time, the Huai River natural experiment provides a single instrumental variable, so it is only possible to use the quasi-experiment to estimate a linear relationship between life expectancy and PM concentrations, leaving unanswered questions about whether the relationship is nonlinear.

Panel C. Distribution of gains in life expectancy from pollution reduction

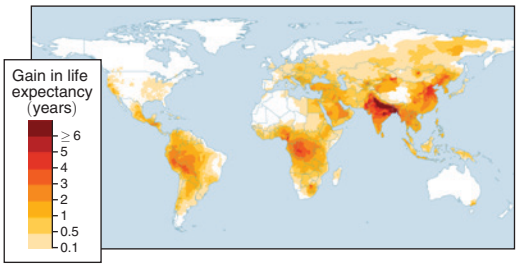


FIGURE 5. AIR POLLUTION CAUSES SIGNIFICANT AND HETEROGENEOUS LOSS OF LIFE (*continued*)

Notes: Panel A shows fitted values from a local linear regression of life expectancy on distance from the Huai River estimated separately on each side of the river, taken from Ebenstein et al. (2017). Panels B and C comes from Greenstone and Hasenkopf (2023). Panel B compares the average per capita life years lost by cause: air pollution estimates are from Greenstone and Hasenkopf (2023), impacts of other causes of mortality are from Global Burden of Disease and WHO Life Tables. Panel C shows the global distribution of the potential gain in life expectancy that could be achieved by reducing air pollution to the WHO standard ($5\mu g/m3$) from AQLI (Greenstone and Hasenkopf 2023). Potential gains in life expectancy are calculated by multiplying the estimated loss in life expectancy from $PM_{2.5}$ from Ebenstein et al. (2017) by the difference between local pollution levels and the WHO standard.

Figure 5, panel B uses the AQLI data and finds that air pollution is the greatest external threat to human well-being on the planet. This is not an exaggeration. The average person on the planet is losing more than two years of life expectancy due to particulate air pollution. That's roughly equivalent to smoking and much more than the losses of life expectancy due to alcohol and drug use, unsafe water, road injuries, HIV, malaria, and conflict and terrorism.

This headline number, however, hides substantial heterogeneity in air pollution and consequently its mortality burden. Figure 5, panel C is a map of the world and depicts the estimated gain in life expectancy from reducing PM air pollution to the WHO standard with darker colors indicating larger potential gains. It is geographically resolved to the level of 47,000 regions around the world that correspond to counties in the United States, districts in China and India, and comparable jurisdictions elsewhere. The average loss in life expectancy from exceeding the WHO standard varies greatly; it is just 0.3 in the United States but 2.5 and 5.3 in China and India, respectively.

It is apparent that air pollution is a great health risk, especially compared to the attention it receives, and that governments have a very mixed record of controlling it. Government policies have led to great reductions in several countries and regions (e.g., the United States, Europe, and Japan) over the last several decades, and in the most recent decade, China has made substantial progress. However, PM pollution continues to cause billions of people around the planet to lead sicker and shorter lives. There is even emerging evidence that early life exposure affects adult earnings (Isen, Rossin-Slater, and Walker 2017; Ebenstein and Greenstone 2023), potentially through compromising cognitive abilities.

Returning to the global energy challenge, these findings make clear that the air-pollution-related externalities from the use of fossil fuels are substantial. A solitary focus on the private costs of energy sources understates their full or social costs.

Fact 6: Fossil Fuels Cause Climate Change and Its Large and Unequal Projected Impacts

Air pollution is not the only externality associated with fossil fuels. GHGs are the other primary one. Fact 6 is that the use of fossil fuels leads to GHG emissions that are causing the climate to change rapidly. The projected impacts are substantial and heterogeneous, with the largest projected impacts occurring in parts of the world that today are hot, poor, or both. These impacts are a second reason that the social costs of fossil fuels exceed their private costs.

Figure 6, panel A provides a recent estimate from the Rhodium Group of where the world is with respect to climate change (Larsen et al. 2023). The left part of the figure reports historical emissions of GHGs, accounting for all gases, in billions of metric tons of CO_2 equivalent (CO_2e) with probabilistic projections out to 2100. In 2022, emissions were 51 billion metric tons of CO_2e, and the mean projection is for this to decline to 40 before increasing again to 44 at the end of the century. The projected decline in the coming decades reflects projected emissions reductions in the power and transportation sectors for the next several decades due to the expansion of wind, solar,

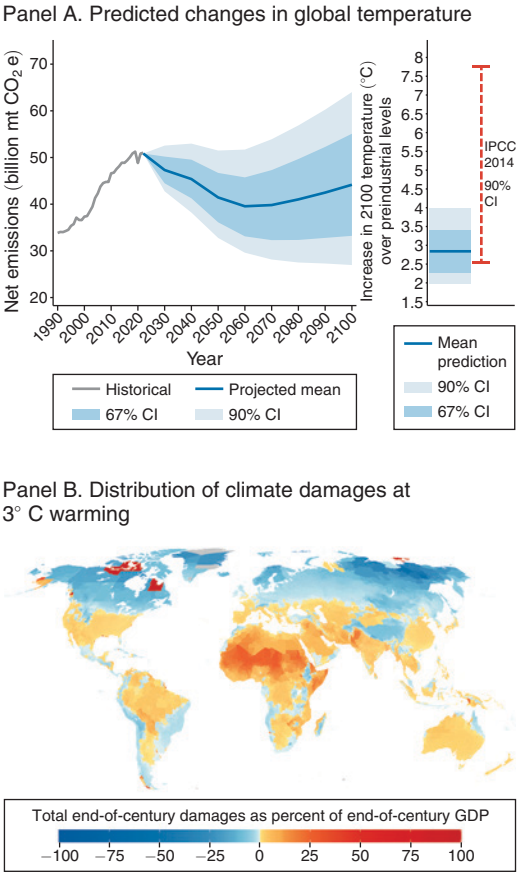


Figure 6. Climate Change Will Cause Heterogeneous Loss of Life

Notes: Panel A shows predicted emissions and change in global temperatures from the 2023 Rhodium Climate Outlook (Larsen et al. 2023). On the left-hand side, net emissions are billion metric tons of CO_2 equivalent, including removals. The right-hand side includes a marker for the prediction of increase in global temperature from IPCC (2014). Panel B shows predicted end-of-century damages as a percent of projected end-of-century GDP from ongoing CIL work (Jina et al. 2024). End of century damages as a percent of end-of-century GDP are calculated as the sum of damages from 2080–2099 divided by the sum of GDP over that period. Estimates come from a model accounting for the benefits of adaptation and income growth. All values shown are based on a scenario with 3.0° C of warming at end of century and are reported separately for 25,000 regions that together comprise the world.

and electric vehicles. However, the projections also indicate that, later in the century, increases in demand in these sectors exceed the declines in carbon intensity such that emissions begin to increase again by 2070.

The right part of the figure reports the temperature increase by 2100 relative to preindustrial temperatures. The Rhodium mean projection is that the world is on a path for global temperatures to increase by 2.8° C (5.0° F) with a 90 percent confidence interval of 2.0° to 4.0° C (3.6° to 7.2° F). This is a marked change from shortly before the 2015 Paris Agreement when the 90 percent confidence interval was 2.5° to 7.8° C (4.5° to 14.0° F) (IPCC 2014). This reduction in projected warming is due to several factors including unexpected declines in the cost of solar, wind, and batteries, new policies, and a reduction in climate scientists' assessment of the likely temperature increase in response to a given increase in greenhouse accumulation in the atmosphere (i.e., the climate sensitivity). It is evident that there has been meaningful progress in slowing temperature change, but we are still headed to a substantially warmer world over a very short geological period.

Figure 6, panel B is based on research that I am conducting with my colleagues at the Climate Impact Lab (CIL) to quantify the damages from warming (Jina et al. 2024). This research is ongoing, so the results should be considered preliminary. The figure plots the damages due to warming as a percentage of GDP at the end of this century in a scenario with 3° C end-of-century warming; blue indicates gains (i.e., climate change is beneficial), and red represents losses.[17] The damages are reported separately for each of approximately 24,000 regions that together comprise the entire planet; they correspond to counties in the United States and to districts in China and India (see Carleton et al. 2022 for details). These total damages are calculated as the sum of damages from five sectors reported in separate papers: the papers provide projections of climate change's impact on mortality due to temperature change (Carleton et al. 2022), energy consumption (Rode et al. 2021), agricultural production (Hultgren et al. 2022), sea level rise and storm damages (Depsky et al. 2023), and the disamenity value of exposure to extreme temperatures while working (Rode et al. 2022).[18] These papers provide detailed

[17] End-of-century damages as a percentage of end-of-century GDP is calculated as the ratio of total damages from 2080–2099 to total GDP from 2080–2099.

[18] Nath et al. (2024) develop the first estimates of the social cost of a marginal ton of carbon dioxide emissions

projections of climate change impacts in these sectors but undoubtedly are missing several other sectors where climate change is likely to influence human well-being (e.g., migration) and any cross-sector interactions.

The map illustrates the striking heterogeneity in projected damages across the planet. There are many areas, particularly those nearer to the equator, where projected damages are substantial, because they already have hot climates and tend to be relatively poor by global standards. For example, Lahore, Pakistan, is expected to be among the hardest hit places with projected end-of-century damages that are roughly equal to 14 percent of end-of-century GDP. In contrast, there are wide stretches of the planet where climate change is projected to be beneficial on net, with most of those places in Russia, Scandinavia, and Canada. These locations are projected to benefit because they have such cold and dangerous winters today and warming will alleviate them. Oslo, Norway, is an interesting contrast to Lahore: climate change is projected to produce end-of-century benefits roughly equal to 9 percent of end-of-century GDP.

This heterogeneity across space is even more complicated because the areas that are projected to benefit are lightly populated and the places with projected damages are home to several billion people. In a world without borders, large-scale migration from the red areas to the blue ones would be a natural form of adaptation. However, borders matter. Time and time again, countries worldwide have demonstrated a limited appetite for receiving even a fraction of the immigration it would take to adapt to climate change. Nevertheless, it seems apparent that there will be significant climate-induced demand for migration in the coming decades. The exact geopolitical consequences and costs of such demand are difficult to predict and certainly not captured in Figure 6, panel B.

Table 1 summarizes the data underlying Figure 6, panel B, considers alternative climate scenarios, and reports on the SCC or the monetary damages associated with the release of an additional metric ton of CO_2. The columns correspond to the entire world, the 38 OECD countries, the 28 countries where per capita income is currently less than $1,085 that are therefore labeled low income by the World Bank, and the remaining 153 countries. The 2021 average per capita income (in 2023 US dollars) among the 1.4 billion people living in the OECD countries is $48,830. The 700 million people in low-income countries have an average per capita income of $810 or about 1.5 percent of the OECD level. The remaining 5.8 billion people have a 2021 average per capita income of $6,830.

Panel B reports end-of-century estimated mean local damages as a percentage of end-of-century local GDP for the average person. The 90 percent confidence interval of these estimated projections is reported below in parentheses, where the uncertainty reflects climate and econometric uncertainty (Jina et al. 2024). Globally, population-weighted mean local end-of-century damages as a percent of local GDP are 1.1 percent, 2.4 percent, and 5.0 percent for end of century temperature changes of 2.0° C, 3.0° C, and 4.0° C, respectively. Within OECD countries, the projected damages for the average person are minimal at −0.6 percent (i.e., a gain of 0.6 percent), 0.0 percent, and 0.3 percent of local GDP at the 3 end-of-century temperature changes, respectively. The 90 percent confidence intervals are relatively tight. The analogous estimated damages for the "Rest of World" category are 1.1 percent, 2.4 percent, and 4.9 percent of GDP with larger confidence intervals. Finally, the estimated projected damages in today's poorest countries are substantially larger (2.0 percent, 3.8 percent, and 8.1 percent). Additionally, the associated confidence intervals are quite wide, suggesting meaningful probabilities of very large damages in these countries.[19]

I take three important points from these results. First, the inequality in climate damages is clear. The places with the largest damages as a percentage of GDP are today's poorest places. In contrast, today's richest places have the smallest projected damages. Second, damages increase nonlinearly with temperature largely due to

that combine theoretical insights on the welfare economics of climate change with a rich set of empirically grounded, probabilistic projections of future climate damages. The damages are from these five sectors.

[19] For example, CIL analysis from preliminary and ongoing work in Jina et al. (2024) suggests that the probability of population-weighted average local damages as a percent of end-of-century local GDP exceeding 10 percent (20 percent) at 3.0° C is 10 percent (1 percent) and at 4.0° C is 39 percent (5 percent).

Table 1—Impacts of Climate Change

	Global (1)	OECD (2)	Low income (3)	Rest of world (4)
Panel A. Demographic characteristics				
Population (billions)	7.9	1.4	0.7	5.8
GDP per capita (US$)	$19,590	$48,830	$810	$6,830
Panel B. End-of-century local damages (percent of end-of-century local GDP)				
2° C warming	1.1	−0.6	2.0	1.1
	(−2.3, 5.8)	(−2.7, 1.6)	(−2.4, 6.5)	(−2.4, 6.1)
3° C warming	2.4	0.0	3.8	2.4
	(−2.8, 9.4)	(−3.1, 3.8)	(−2.3, 12.6)	(−3.1, 10.2)
4° C warming	5.0	0.3	8.1	4.9
	(−3.2, 13.8)	(−4.1, 4.6)	(−2.5, 21.4)	(−3.6, 14.4)
Panel C. Social cost of carbon (US$)				
2° C warming	$75	—	—	—
	(−$36, $183)			
3° C warming	$387	—	—	—
	($134, $632)			
4° C warming	$1,060	—	—	—
	($281, $1,972)			
US EPA (2.8° C warming)	$225	−$13.5	$221.6	$16.9

Notes: Income group populations and GDP per capita were obtained from World Bank (2024d) and reflect 2021 values. Damage and SCC temperature bins indicate the degrees of warming by end of century relative to preindustrial levels. End-of-century damages are obtained from ongoing and preliminary CIL analyses combining damages from agriculture, mortality, energy, coastal and sea-level rise, and labor disutility across roughly 25,000 impact regions (Jina et al. 2024). End-of-century local damages as a percent of end-of-century local GDP are calculated as the sum of damages from 2080–2099 divided by the sum of GDP over that period for each impact region. Displayed damage values are population-weighted means across impact regions, with 90 percent confidence intervals shown in parentheses. Confidence intervals are based on the distribution of damage estimates across 1,000 model runs, with resampled climate and economic projections and dose-response function coefficients. SCCs are calculated at each warming level based on methodology from Jina et al. (2024) and Nath et al. (2024). The disaggregation of the SCC by country income groups is from a back-of-the-envelope calculation obtained by determining the portion of the net present value of cumulative climate damages at the end of century faced by each income group and attributing that portion of the overall SCC to the income group. US EPA SCC taken from Carleton and Greenstone (2022) and converted to 2023 US dollars.

nonlinearity in the damage function. These estimates leave little doubt that there are large (gross) benefits from avoiding the higher end of potential climate change. Third, the uncertainty in these estimates is substantial, especially outside the OECD. With standard levels of risk aversion, individuals would be willing to pay a significant amount just to remove the uncertainty around damages.

Finally, panel C presents preliminary estimates of the SCC from ongoing work with my colleagues at the CIL. The SCC is the present value of the monetary damages associated with the release of an additional ton of CO_2, accounting for the long periods that they will remain in the atmosphere and influence the climate and human well-being. The SCC is central to setting the stringency of US government regulations that restrict GHG emissions (EPA 2023; Carleton and Greenstone 2022). Additionally, the US value is influential in other settings because it is the most closely tied to the frontiers of economics and climate science; it has been adopted by several countries (e.g., Canada and Mexico), states (e.g., California and New York), and corporations.

The reported estimated value of the SCC is $387 (in 2023 US dollars) when the estimated end-of-century increase in temperature is 3.0° C (approximately equal to the mean temperature increase projection in Figure 6, panel A).[20] It

[20] The values of the SCC at each warming level are based on combining the approaches of Jina et al. (2024) and Nath et al. (2024)

is smaller ($74) at 2.0° C and larger ($1,060) at 4.0° C. Converting to more familiar metrics, these values of the SCC imply that a gallon of motor gasoline is projected to cause damages equal to approximately $0.50 in the 2.0° C scenario, $3.50 in the 3.0° C scenario, and $9.50 in the 4.0° C scenario. The rapid increase in projected damages from a given increase in GHG emissions with temperature increases is largely due to the convex damage functions.

The bottom row of panel C implements all the assumptions embedded in the US EPA approach in its 2023 update of the SCC,[21] including a mean temperature increase of 2.8° C (EPA 2023). The SCC with the EPA's full set of assumptions is $225 (in 2023 US dollars).[22] For this exercise, it is additionally possible to geographically break out the projected climate damages due to an additional ton of CO_2 emissions. Specifically, under the back-of-the-envelope assumption that SCC contributions are proportional to cumulative climate damages at the end of the century, the CIL work finds that the marginal ton of CO_2 produces small benefits ($13.5) in OECD countries that are vastly outweighed by large losses in the remainder of the world ($238.5).

Finally, it is noteworthy that the SCC can be used for a green GDP style accounting calculation. Global GDP was $101.3 trillion in 2022, but this production came at the cost of bequeathing to ourselves, our children, their children, and so on roughly $11.5 trillion (= EPA SCC × global CO_2e metric tons of emissions) of climate damages.[23] Thus, 2022 global GDP would be about 11 percent lower if it accounted for the projected damages associated with GHG emissions.[24]

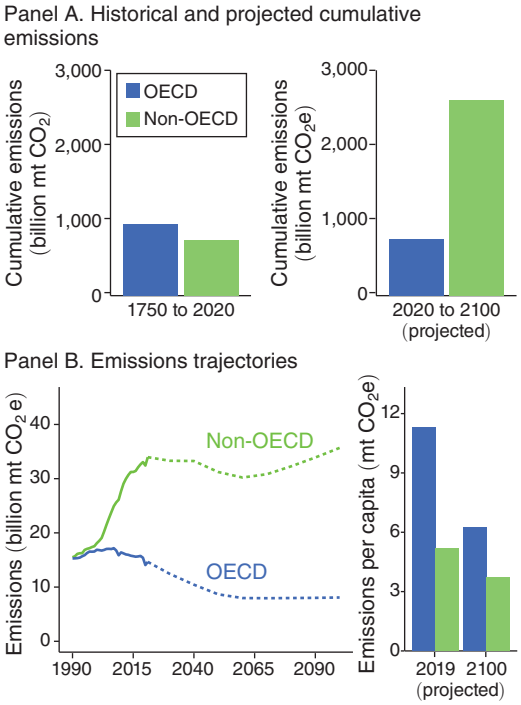


FIGURE 7. POLITICALLY CHALLENGING GHG ACCOUNTING

Notes: Panel A shows historical and projected cumulative emissions. Historical emissions are collected from Our World in Data. Projected emissions are collected from Larsen et al. (2023). Panel B shows historical and predicted emissions and emissions per capita from Larsen et al. (2023). In time series predictions are marked with dotted lines, and per capita emissions shown in 2019 and 2100.

Fact 7: Climate Arithmetic Is Cruel

Fact 7 is that the global energy challenge exposes climate change's cruel arithmetic. For the OECD and non-OECD groups of countries, Figure 7, panel A reports their cumulative emissions of CO_2 in metric tons since the Industrial Revolution and their projected emissions of CO_2e between 2021 and 2100. Through 2020, OECD countries accounted for 955,151 billion metric tons of CO_2, which is roughly 56.2 percent of emissions to date; today's wealthy countries are responsible for a majority of historical emissions and a disproportionate share when

[21] Several of these assumptions differ from those in Jina et al. (2024) and Nath et al. (2024). The sensitivity of the results to each of the different assumptions is beyond the scope of this lecture but will be detailed in Nath et al. (2024).

[22] This is equal to the $190 (in 2020 US dollars) that the EPA cites (EPA 2023).

[23] Carbon dioxide equivalent or CO_2e calculations are more comprehensive than CO_2 ones because they account for CO_2 emissions *and* other GHGs (e.g., methane, nitrous oxide, and ozone) measured in metric tons of CO_2. The other GHGs are converted into metric tons of CO_2 emissions by scaling them to have the same global warming potential as one metric ton of CO_2.

[24] It is reasonable to consider the planet's emissions in a single year as marginal because of the long half-life of GHGs in the atmosphere (e.g., carbon dioxide's half-life is more than a century). For example, there were 50.6 billion metric tons of CO_2e emissions in 2022, which is a small fraction of the 1.96 trillion metric tons of carbon dioxide released since the beginning of the Industrial Revolution in 1790 (Larsen et al. 2023).

one accounts for their share of global population (17.3 percent in 2020). The projections for the remainder of the century tell a very different story: OECD countries are projected to account for another 746,192 billion metric tons of CO_2e, but the non-OECD countries' projected cumulative emissions are much larger at 2,599,515 billion metric tons.

Figure 7, panel B reports historical and projected emissions, both gross and per capita, which elicit several related observations (Larsen et al. 2023). Fairness would suggest that the non-OECD countries have a right to greatly increase their emission so that people around the globe are treated equitably. This is both because today's rich countries are responsible for a disproportionate share of historical GHG emissions and because of the large current differences in per capita emissions. However, the planet and its atmosphere only care about total emissions and have no interest in history or equality or any other metric.

Basic arithmetic then means that significantly restraining climate change would require extraordinary emissions cuts from the non-OECD countries. To provide some quantitative context, restraining expected changes in temperature to 2.0° C is often proposed by scientists as an important goal (UNFCCC 2010). A standard projection of how to achieve this goal would, for example, require a reduction of roughly 38 billion tons of CO_2e in 2100 relative to current projections. Today's OECD countries are only projected to emit 8 billion tons of CO_2e at the end of the century, while non-OECD countries are projected to emit 36 billion tons. So even if the OECD countries become carbon neutral by then, the non-OECD countries would only be left with 6 billion tons of emissions if the 2.0° C target is to be hit. Put another way, meeting the 2.0° C target requires the non-OECD countries to cut their end-of-century CO_2 emissions by roughly 85 percent relative to their current baseline projections.

This example illustrates the cruelty of climate arithmetic. Specifically, restraining climate change to the temperature increases recommended by scientists requires the vast majority of emissions reductions to come from today's poorer countries. These countries are most in need of higher energy consumption (and least willing to pay extra for low-carbon energy sources), account for the great majority of the world's population, and have emitted the least to date. It is not difficult to imagine that they have an incredulous reaction when they are confronted with unfunded calls from the international community to reduce their GHG emissions. At the same time, these poor countries are projected to suffer the most absent emission cuts, because their projected damages are so much higher. These countries are between a rock and a hard place and this is the cruelty of climate arithmetic.

B. *The Consequences of the Global Energy Challenge*

These seven facts paint a stark picture, defining the global energy challenge and the attendant trade-offs that every society faces. They also highlight that the reason this is such a difficult challenge is that these trade-offs differ so much around the world, making it unlikely that, for example, countries can agree on a common price for carbon. Yet the fact that GHG emissions have the same effect no matter where they are emitted means that each society's welfare depends on all other societies' actions.

The United States is a poignant example of the trade-offs that the global energy challenge introduces. Energy is generally inexpensive and reliable. The average American is wealthy by global standards with a per capita income of about $76,000 (World Bank 2024b) and a consumption of about 13,000 kWh of electricity annually. Additionally, air pollution concentrations are low both in comparison to the United States' past and the rest of the world today, largely due to the success of the Clean Air Act over the last half century (Chay and Greenstone 2005; Currie and Walker 2019). With respect to GHG emissions, the last two decades have seen meaningful reductions (e.g., roughly 14 percent between 2005 and 2021); however, much of this decline has been unrelated to policy and happened for arguably exogenous reasons. Specifically, innovation has reduced the costs of recovering natural gas (due to advances in hydraulic fracturing) and of renewables, and, together, this has led to large reductions in coal's market share in power generation.[25]

[25] In modern power plants, coal combustion emits roughly twice as much CO_2 as natural gas per unit of electricity produced. Ignoring the need for backups, renewables are a zero-CO_2 energy source.

Thus, the United States is in a strong position with respect to inexpensive and reliable energy and clean air, but it has struggled with making policy commitments to GHG emissions reductions that are consistent with balancing benefits and costs or the targets advocated by scientists (e.g., limiting temperature increase to 2.0° C). In making this point, I do not ignore the important roles of federal regulations to restrict GHG emissions via renewable portfolio standards (Greenstone and Nath 2024), renewable production and investment tax credits, and recent policies enacted by the Biden Administration through the Inflation Reduction Act. However, the United States is still on track to emit 3.6 billion metric tons of CO_2e in 2050 compared to 6.2 billion metric tons—about 18.8 metric tons per capita—in 2021 (Larsen et al. 2023). Overall, even in a wealthy country like the United States that has inexpensive and reliable energy and clean air, the willingness to spend more on energy today in exchange for a reduction in climate risk has been modest to date.

India is another especially useful example, as it demonstrates that the global energy challenge's trade-offs vary around the world. Annual per capita electricity consumption was a little more than 1,200 kWh in 2021 (2021 per capita income was $2,238), air pollution concentrations are very high (the average Indian would live about five years longer if air pollution were brought into compliance with WHO standards), GHG emissions are about 3.9 billion metric tons of CO_2e total (or just 2.8 metric tons per capita) and are projected to grow rapidly in the coming decades, and climate damages are projected to be among the largest on the planet (approximately 13.1 percent of GDP at the end of the century). Returning to Figure 1, panel B, which illustrated that extremely hot days are roughly ten times deadlier in India than in the United States, it seems likely that India's focus will be to increase incomes so they can move from the green line's hardships to safer lives on the blue line. This requires inexpensive energy, which generally means fossil fuels and their accompanying emissions of local air pollutants and GHGs. Indeed, the Indian government said in December of 2023 that it plans to double coal production by 2030 (Rathi 2024). This suggests that to keep energy inexpensive and reliable, India is willing to accept higher levels of air pollution and greater global climate damages.

These two examples illustrate that the global energy challenge involves countries worldwide seeing the same problem of balancing inexpensive and reliable energy, air pollution, and climate risk very differently and consequently making very different choices. These choices would not merit such careful consideration and could be entirely a domestic issue, except that GHGs are a global pollutant that does not respect borders. Put plainly, the United States is greatly affected by India's WTP today to reduce GHG emissions, just as India is greatly affected by the United States' WTP for GHG reductions.

The particularly biting part of this interdependence is that a big part of confronting climate change requires large reductions in GHG emissions in countries that are relatively poor and/or have emitted relatively little historically. In general, these countries do not see it as in their interest to pay much extra for energy today to reduce their GHG emissions, nor is there much political momentum in OECD countries to cover these extra costs by, for example, subsidizing low-carbon energy consumption in India, Bangladesh, China, or other non-OECD countries. Indeed, I am unaware of any evidence that a promise to subsidize energy in other countries would help anyone get elected to the US Senate, even from a state like California that has such a rich history of leading on environmental policy!

With these heterogeneous trade-offs and interdependencies, is there a way to make the global energy challenge easier to manage? The answer is yes. But this involves characterizing the many market failures that plague energy markets around the world, identifying potential solutions that are socially beneficial for the countries making them (not for other countries), and testing whether policy reforms deliver the intended results. This is the subject of the next section.

II. Energy Market Failures and Some Solutions

This section considers four categories of market failures that are endemic to energy markets around the world and make confronting the global energy challenge more difficult.[26] Here,

[26] See Greenstone and Jack (2015) for an analysis of the role of market failures in determining environmental quality in developing countries.

Table 2—Private, External, and Total Social Cost per kWh for Electricity Plants, United States (2028)

Source	Share of total generation	Private costs	Particulate matter costs	GHG costs	Total social cost
	(1)	(2)	(3)	(4)	(5)
Panel A. Current baseload					
Coal	19.6	7.6	1.3	28.5	37.3
with 90% CCS		12.1	1.3	2.8	16.2
Natural gas	39.0	4.3	0.1	14.8	19.1
Nuclear	17.9	10.6	0.0	0.4	11.0
Panel B. Renewables with NGCT backup					
Solar	4.7	6.8	0.0	5.0	11.8
Wind (onshore)	10.1	7.0	0.0	3.8	10.8
Wind (offshore)	0.0	14.3	0.0	3.1	17.4
Panel C. Renewables with battery backup					
Solar	4.7	9.2	0.0	1.4	10.6
Wind (onshore)	10.1	9.2	0.0	0.5	9.7
Wind (offshore)	0.0	15.7	0.0	0.7	16.4

Notes: This table displays levelized private, particulate, and GHG costs for US plants coming online in 2028 in ¢/kWh (2023). Coal (90 percent CCS) is a coal plant with CCS technology capable of capturing 90 percent of carbon emissions. Column 1 displays each source's share of total electricity generation in 2022. For renewables with backups, only generation from the renewable is considered. Not all generating sources are included in the table, so shares do not sum to 100. Column 2 is the private LCOE. It is the cost of energy such that the sum of present-discounted annual revenue is equal to the sum of present-discounted annual private costs over the source's lifespan. Annual costs and generation are computed using inputs from EIA (2023). Column 3 is the mortality cost due to particulate emissions and is equal to the ratio of mortality damages to total energy generation. It is computed at the level of the source category. Annual mortality damages from particulates are the sum of damages across census tracts. Damages within a census tract are equal to the product of the change in ambient $PM_{2.5}$ concentrations due to generation from sources in the source category, the tract's population, a mortality concentration-response function, and the value of a statistical life year (VSLY). $PM_{2.5}$ concentration data for coal and natural gas sources come from Hernandez-Cortes, Meng, and Weber (2023), population from Manson et al. (2023), the concentration-response function from Ebenstein et al. (2017), and the VSLY is derived from the EPA's suggested VSL. Column 4 is equal to the present-discounted sum of annual GHG damages divided by the sum of present-discounted annual generation. Annual GHG damages are lifecycle CO_2e emissions per kWh multiplied by annual generation and scaled by the SCC. Lifecycle emissions come from Nicholson and Heath (2021). The SCC comes from EPA (2023). Column 5 sums columns 2–4.

I will outline them and discuss approaches to correcting them.

A. *Market Failure 1: Energy Prices Fail to Reflect Social Costs*

The first market failure is that energy is rarely priced at its full social cost. This mis-pricing affects many energy sources, but it is most acute when it comes to fossil fuels. As the last section made clear, the two large externalities associated with fossil fuels are that their combustion to produce energy involves the release of airborne PM (or its antecedents) and GHGs. These externalities are generally not priced, so fossil fuel users can harm others without any penalty. The result is that fossil fuels compete with other energy sources on a playing field tilted to their advantage.

Table 2 provides an opportunity to assess the full social costs of alternative electricity sources in the United States and to consider how choices would differ if energy prices reflected their social costs. Column 1 reports the sources' share of electricity generation in 2022 in the United States. Columns 2–5 present several projected per kWh costs for new power plants that come online in 2028.[27] Column 2 repeats the LCOEs that were the basis of Figure 4 panel B (recall that these are national averages, and costs will vary across location). Column 3 reports the mortality costs due to PM emissions per kWh,[28] and

[27] It takes about 4–5 years to build a nuclear plant, so I use 2028 as a means of comparing all technologies in the same year.

[28] See the table notes for details on this calculation, including the choice of the elasticity of life expectancy with

column 4 details the monetary value of damages per kWh from GHG emissions, computed by multiplying CO_2e emissions by the EPA's value of the SCC ($225 in 2023 US dollars). Finally, the estimated total social costs are reported in column 5 and they are the sum of private, PM, and GHG costs. This last column can be used to consider a world where the playing field is level and the costs of all energy sources reflect their full social costs.

Column 3 provides an opportunity to assess the magnitude of the PM externalities associated with each electricity source. The PM costs from a US coal plant are approximately 1.3 cents per kWh and just 0.1 cents per kWh from a natural gas plant. They are effectively zero for all the other technologies. At these relatively low levels, the pricing of the PM externality would further drive coal out of the money, likely making the renewables with both forms of backup more competitive in several locations. The relatively small magnitude of the PM externality reflects the great progress that Clean Air Act regulations have made in reducing emissions per unit of electricity and output in the United States over the last several decades (Currie and Walker 2019; Shapiro and Walker 2018). As a point of comparison, the PM externality costs from coal are several times higher in the many parts of the world with much less stringent air pollution regulations.[29]

Columns 4 and 5 reveal that pricing GHG emissions at the US government's SCC would dramatically reshape US electricity markets. In practice, it would cause the two fossil fuel sources to be substantially more expensive than nuclear, solar, and onshore wind.[30] The increase in coal prices is especially noteworthy, as the GHG costs are almost four times the private costs. Overall, these estimates make clear that it would be disingenuous to be surprised about the continued dominance of fossil fuels in the electricity sector because we have designed electricity markets to choose them.

The most efficient way to internalize the GHG costs of alternative energy sources is through carbon pricing via a CO_2 tax or cap-and-trade program. Since CO_2, unlike PM, is a global pollutant, it is informative to compare the state of global carbon pricing to estimates of the climate damages caused by CO_2 emissions. According to the World Bank, the average metric ton of CO_2 currently faces a price of about $6, which is much less than the US government's SCC of $225. This low price reflects that approximately 77 percent of the world's CO_2 emissions are unpriced and that the remaining emissions are priced at an average of roughly $24 per metric ton. Importantly, the implied global price of CO_2 would be even lower if the net effect of other policies, importantly including fossil fuel subsidies, were included (Gro Intelligence 2024).

The politics of carbon pricing are complicated in the United States and elsewhere, but two of the most frequent criticisms can be readily addressed with design features. First, the share of income devoted to energy is decreasing in income, so simple carbon pricing is likely to be regressive. This could be remedied by taking a portion of the significant revenue generated by carbon pricing and rebating it to the lower parts of the income distribution to blunt the regressivity (Rosenberg, Toder, and Lu 2018).[31] Second, carbon pricing would disadvantage US industries with heavy emissions if other countries do not price carbon emissions within their borders. Border tariff adjustments are a standard way to protect domestic industry, and the EU has indeed begun to implement such tariffs. The broader point is that some of the political problems with carbon pricing can be readily addressed through policy design.

Before closing this subsection, it is important to underscore that the current structuring of energy markets to focus on private, not social, costs does not result from a law of physics or a religious edict; we choose these

respect to $PM_{2.5}$ and the value of a statistical life (VSL).

[29] For example, a similar methodology suggests that the PM costs of electricity from a coal plant in India are 9.3 cents per kWh, even when using a lower VSL based on the assumption that the elasticity of the VSL with respect to income is one.

[30] The leakage rates of methane in the recovery and transportation of natural gas are a topic of considerable debate. Here, I have very conservatively (and almost certainly incorrectly) assumed zero leakage rates, as the qualitative finding that natural gas would be more expensive than several alternatives if GHGs were priced at the SCC is unchanged. Alvarez et al. (2018) estimate a methane leakage rate of 2.3 percent, which if priced into the cost of natural gas, would make natural gas even less competitive with alternative generation sources.

[31] See Currie and Gahvari (2008) and Banerjee et al. (2024) on the economics of social transfers.

market structures, and therefore we choose their consequences, even though they often deliver socially suboptimal outcomes.

There are several candidate explanations for these seemingly perverse policy choices, and here I draw attention to four especially salient ones. The first explanation is a standard political economy one. Structuring energy markets to choose energy sources based on social costs (e.g., by pricing CO_2 and PM), rather than based on private costs only, would have diffuse benefits spread over the entire population. However, the costs would be largely concentrated in a few industries, their workers, and the locations that host them (Stiglitz 1998). In these settings, the group with the most to gain (or lose) often gets their way politically through lobbying and other approaches to applying political pressure. Second, state capacity is low in many countries, and implementing and executing corrective taxes and regulations is often unsuccessful (see, e.g., Duflo et al. 2013; Ryan 2021). Third, the impacts of air pollution and climate change affect people unequally—with the largest costs often borne by the poor both across and within countries (Pande 2023). However, the rich and powerful often have disproportionate influence on policy, and this can lead to policy choices that benefit them over the greater good. Fourth, there is a growing body of evidence that fossil fuel companies have engaged in disinformation campaigns to obscure the facts about environmental externalities (e.g., "Clean Coal") that go beyond the standard lobbying covered by the first explanation (Oreskes 2011).

B. *Market Failure 2: Too Little Innovation*

Innovation is generally a public good, and therefore its supply is likely to be suboptimally low without public support. Technical innovation and policy innovation are two especially important forms of innovation for making progress on the global energy challenge.

Technical innovation offers the promise of reducing the costs of energy, but it is unlikely we are getting the socially optimal level of such innovation, especially among low-carbon energy technologies. This market failure stems from the fact that firms that conduct R&D, particularly in early phases, cannot capture all the benefits because some of the ideas will spill over to other firms. The consequence is that privately optimal R&D levels are below the socially efficient level precisely because, in making their R&D decisions, firms do not account for benefits to other firms. A solution to this classic market failure is to subsidize technical R&D. Below, I discuss why this might be an especially important issue for low-carbon technologies.

The second innovation problem, which in my judgment has received insufficient attention, is that there is not enough innovation in policy design and execution. Like the technical innovation problem, this one results from a difference between the benefits for individual actors, which in this case are governments instead of firms, and benefits to society, which are potentially countries around the globe. Specifically, policy innovation is costly for governments in terms of resources and time, as well as risky for civil servants' career advancements. However, the benefits of policy innovation—for example, better understanding the impacts of novel policy instruments—are not limited to the jurisdiction where the experimentation took place, often extending to many other jurisdictions in the same country and across international borders. The result is that individual governments are likely to underinvest in policy innovation from a social perspective.

This subsection discusses these two market failures and ways to address them in greater detail.

Technical Innovation.—Both theory and empirical evidence suggest that it is likely that there are socially suboptimal levels of "basic" research.[32] It is standard to assume that firms make their investment decisions based on the returns that accrue to them and generally do not consider any spillover benefits to other firms or the overall economy. In the presence of such spillovers, their decisions are privately optimal *and* socially suboptimal (especially for

[32] I thank John Deutch for burning into my brain that the public good problem is likely to not just be limited to "basic" research. In particular, there is likely to be suboptimal investment in the demonstration of new technologies at scale, especially in the energy sector (e.g., carbon capture and storage or advanced nuclear, like small modular reactors) where the necessary investments are very large. In summary, the demonstration of the feasibility of new, and especially expensive, technologies is a public good and can benefit many companies.

early-stage "basic" research and demonstration of new technologies). Recent research supports the presence of such spillovers. Specifically, Bloom, Schankerman, and Van Reenen (2013) find that gross social returns to R&D are about 55 percent, which is more than double their estimated private returns of 21 percent.

In the case of low-externality or externality-reducing energy technologies, there may be additional reasons to publicly subsidize R&D. First, as discussed above, the political barriers to pricing externalities are great even when the benefits are almost entirely local (e.g., particulates air pollution). Thus, it is possible that spending beyond the efficient level on R&D on "clean" technologies may be justified as a second-best policy for shrinking the difference in the private costs of production between high- and low-externality energy sources.

Second, it may be welfare enhancing but politically infeasible for rich countries to pay poor countries to reduce their GHG emissions. For example, rich countries may find that the reduction in climate damages in their countries exceeds the costs of paying poorer countries to reduce their emissions, and thus such transfers would be beneficial on net. However, such transparent transfers are unlikely to be popular politically. An alternative that has the potential to achieve the same goal is to provide public support for R&D on low-carbon energy sources. Indeed, successful R&D can reduce the costs for these technologies so that they are adopted globally. Although less direct than transfers, this second-best rationale has the potential to reduce global climate damages, including for the countries that fund the R&D.

Third, Acemoglu et al. (2012, 2016) emphasize that increasing returns to clean technologies that are in competition with "dirty" technologies can justify public subsidies for R&D. These increasing returns may be especially likely in the early years of technologies and/or when they are far from being price competitive with the dirty technologies.

The broad point is that technical innovation is a public good and is likely to be undersupplied by the private market. The classic solution is to provide government support for R&D projects, especially in the early stages of the innovation chain when spillovers are believed to be largest. The design of such policies is prone to political influence, and successful technical innovation policy requires disciplined approaches to ensuring that the spending is not captured or corrupted in ways that are undermining. The US Department of Energy Advanced Research Projects Agency-Energy is often pointed to as a model for avoiding political influence in the energy space, while the National Science Foundation and National Institutes of Health have long traditions of decision making that is guided by science in other areas.

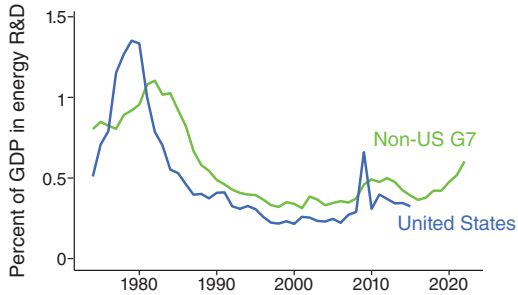


Figure 8. Energy Research and Development Spending

Notes: Lines show energy research and development expenditure as a percent of GDP for the United States and other G7 countries. The G7 countries are Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States. R&D expenditure is from IEA (2023a) with US data available until 2016, and GDP is from World Bank (2024a).

Figure 8 provides an opportunity to assess where the world is on supporting technical innovation in energy by plotting a time series of public spending on energy R&D as a percentage of GDP in the United States and the other G7 countries. This measure includes spending on fossil fuel sources *and* low-carbon energy sources. The high point for these series was in the 1970s in the aftermath of the oil crises. It is difficult to conclude that there has been much of an acceleration in energy R&D spending as awareness of the climate challenge has increased.

For the reasons outlined here, I believe there is a strong economic case for a rapid expansion of funding on technical energy innovation for low-carbon energy sources and complementary technologies. As long as there is a meaningful difference between the private costs of fossil fuels and low-carbon energy sources, many societies, regardless of income level, are likely to continue to rely on fossil fuels that cause global harm through GHG emissions. Thus, a

program to shrink the difference between the costs of fossil fuels and other energy sources has high potential benefits. Additionally, it is at least plausible that it is more politically feasible than alternative policies to reduce emissions.

Policy Innovation.—The road from setting a policy goal to implementing a successful policy is often quite long with many potential roadblocks and dead ends. The role of economists has frequently been limited to identifying market failures and describing theoretical solutions. However, the track record of economists' shouts to "price externalities" and "subsidize innovation" converting into successful policy is not overwhelmingly positive. This is because policy is not made in textbooks or university classrooms but rather in widely varying settings where real people live, real governments govern, key actors' incentives matter, information is far from perfect, distributional impacts influence politicians, and so forth. All too often, the result is inefficient policy, or even outright policy failures, that can undermine trust and confidence in government more broadly.

An attractive solution is relentless experimentation with alternative policy designs, but such policy innovation also faces a public goods problem. The benefits of experimentation are diffuse as many governments, even those in other countries, can reshape their policies based on the results. The costs, however, are concentrated in the jurisdiction that experiments with policy design. These costs can be substantial in time, money, and public employees' attention. Additionally, they can be risky for public employees as negative results can harm their careers.[33] The result is that the social benefits of policy innovation likely exceed the private benefits that accrue to the governments that incur the costs, so we should expect suboptimally low levels of policy innovation. Therefore, there is a strong theoretical case for subsidizing research on policy innovation.

[33] When I worked in the Obama Administration, a colleague led an effort to try out a new policy. The results were not favorable, and it caused some minor political embarrassment for the White House. I vividly remember a senior Administration official asking the employee in a raised voice, "How could you have thought this would be helpful?" My colleague wisely did not answer that it was for policy innovation purposes!

The best path forward for such evidence-based policy innovation has two key features. The first feature is to pay careful attention to what Esther Duflo termed "plumbing" (Duflo 2017), or the specific details, especially the incentives of key actors. These details go far beyond general prescriptions like "price externalities" or "subsidize innovation." Duflo writes "details that we as economists might consider relatively uninteresting are in fact extraordinarily important in determining the final impact of a policy or a regulation, while some of the theoretical issues we worry about most may not be that relevant" (Duflo 2017, p. 1). The decision to engage with these details is often the difference between designing a theoretically beneficial policy on the blackboard and a policy that succeeds in the real world.

The second feature is a relentless commitment to measuring the impacts of policy reforms rather than relying on the all-too-frequent approaches of claiming victory without data or treating implementation as the ultimate outcome. An especially powerful approach to learning about a policy's impacts is through the use of RCTs that rely on true experiments, just as are used in the testing of new drugs, for example. The use of RCTs for policy experimentation dates back at least half a century (e.g., Office of the Assistant Secretary for Planning and Evaluation 1983; Aron-Dine, Einav, and Finkelstein 2013). More recently, this type of policy experimentation was one of the reasons that the 2019 Nobel Prize was awarded to Abhijit Banerjee, Esther Duflo, and Michael Kremer.

This use of RCTs to assess policy reforms has taken on greater momentum in the last two decades, including around questions related to the global energy challenge. Some notable findings include: residential energy efficiency improvements have had a low rate of return in the United States (Fowlie, Greenstone, and Wolfram 2018); performance pay for local tax collectors increased property tax revenues substantially in Pakistan (Khan, Khwaja, and Olken 2016); electronic vouchers were a more efficient means to deliver social assistance than in-kind food assistance to targeted households in Indonesia (Banerjee et al. 2023); biometric smart cards improved the efficiency of delivering employment and pension benefits in India (Muralidharan, Niehaus, and Sukhtankar 2016); the receipt of Medicaid insurance increased

health care utilization, reduced financial strain, and produced mixed impacts on physical and mental health outcomes in the United States (J-PAL Policy Briefcase 2023); the establishment of a pollution market for PM emissions in Gujarat, India, reduced emissions and firms' compliance costs (Greenstone et al. 2024); and machine-learning targeting of inspections increased the frequency that US EPA inspections uncover severe violators (Buchsbaum, Greenstone, and Meckel 2024).

In several cases, the results from these experiments have altered policy in the jurisdictions where they were tested and in other jurisdictions too, underscoring the "basic" nature of the original research. For example, Indonesia rolled out electronic vouchers nationwide to deliver social assistance; the government of Gujarat, India, has substantially expanded the use of pollution markets based on the study's results, and several other Indian states are exploring opportunities to start these markets; the US EPA and the state of Colorado are now using machine learning to target inspections in compliance programs beyond just those in which they were initially tested; and India is now using smartcards to target and deliver social benefits of various kinds throughout the country. In other instances, the study results have not influenced policy to date: the recent expansion of funding for residential energy efficiency upgrades in the United States, despite evidence of their low rates of return, is a powerful example.

Beyond these individual instances, there is emerging evidence on the returns to investing in innovation around social entrepreneurship and social science R&D more generally. A recent paper evaluates the US government's Development Innovation Ventures fund and estimates that benefits were 17 times larger than costs (Kremer et al. 2021). Moreover, the returns were very right skewed. So, just as venture capitalists are willing to invest in many scores of companies with an expectation of finding a handful of Googles or Facebooks, there is a case for expecting and accepting lots of research that doesn't identify promising policy reforms in the name of uncovering a couple "unicorns." Indeed, Greenstone et al. (2024) uncovered that pollution markets in India have benefits that exceed costs by perhaps two orders of magnitude, and, as noted above, this finding is motivating the use of markets in many jurisdictions besides the one where it was tested.

Putting these individual cases and aggregate results together, my judgment is that the net social benefits would exceed costs even if we increased research spending on the global energy challenge substantially. The returns come from learning about what works where it is tested and from other jurisdictions that can act on the results. Moreover, policy innovation is a crosscutting public good that can be useful in addressing all the market failures described in this section (including those described in the next two subsections).

C. *Market Failure 3: Imperfect Information*

A foundational assumption for markets to deliver efficient outcomes is that individuals have perfect, or at least unbiased, information. However, accurate information is a classic public good in that it is rarely in an agent's self-interest to supply the socially optimal level. Thus, just as with innovation, the private market is likely to supply suboptimally low levels. Imperfect information can reduce welfare by leading individuals to fail to optimize their behavior on a wide variety of dimensions. In the case of energy and environment, it can cause individuals to fail to protect themselves from environmental risks, making the consequences of environmental externalities worse, and to fail to take advantage of adaptation opportunities.

Recent research provides examples of just this phenomenon. For example, Baylis et al. (2024) experimentally provided information to residents of Delhi on the health consequences of exposure to air pollution and found that it increases their WTP for reductions in air pollution by more than five times. This finding suggests that individuals' imperfect information caused them to fail to optimally protect themselves from air pollution based on their own preferences. Specifically, individuals would have been more likely to use N95 masks to reduce their exposure to air pollution if they had been provided accurate information about its risks. In a similar vein, Greenstone et al. (2022) find that a 35 percent increase in reported PM concentration in China due to improved monitoring increased online searches for face masks and air filters, which are strong predictors of purchases. In another domain, the experimental provision of more accurate monsoon onset information to rural farmers in India caused them to

update their beliefs and change their planted crops, area under cultivation, and input expenditures (Burlig et al. 2023). Finally, Shrader, Bakkensen, and Lemoine (2023) show that accurate weather forecasts of extreme hot and cold days reduced mortality rates, presumably due to compensatory responses to the forecasts.

Both theoretical and empirical evidence, then, suggest that accurate information is a public good that is undersupplied. Knowledge about the health effects of air pollution and information on climate or weather shocks are two areas where there appear to be particularly strong cases for further public investment. Such funding would allow individuals to better protect themselves from risks and exploit opportunities. More broadly, it would cause them to advocate for policies that better match their preferences.

D. *Market Failure 4: Electricity as a Right*

The final market failure I want to highlight is the idea that electricity is a right rather than a private good.[34] This social norm has gained traction in many parts of the world; its defining feature is that all people deserve electricity regardless of payment. For example, Bolivia's constitution guarantees a right to universal electricity access, and India's Prime Minister wrote, "Everyone has a right to a life of dignity. Traditionally, food and shelter have been seen as the most basic necessities. However, the Modi government has gone beyond this core basket of necessities to include even electricity" (Modi 2019; Burgess et al. 2020). The treatment of electricity as a right sharply contrasts with its treatment as a standard private good in wealthy countries and other parts of the world.

The electricity-as-a-right social norm is seemingly well intentioned, but what are its consequences? Once electricity stops being a private good that must be purchased, nonpayment and illegal grid connections are tolerated. The electricity distribution companies then lose money on every kWh of electricity sold, so they try to sell *less* of their product by limiting access, restricting hours of supply, tolerating load shedding, and so forth (i.e., electricity is not excludable as it is in well-functioning electricity markets), meaning that customers who do not pay receive roughly the same supply as those that do (Burgess et al. 2020).[35] The result is a low-quality, low-payment equilibrium where electricity is not allocated on purely economic grounds.

The market failure here is that mutually beneficial transactions between the distribution companies and their customers cannot take place. Specifically, there are many customers who would be willing to pay more than the costs of supplying electricity, yet these potential transactions cannot take place due to distribution companies' efforts to sell less. The suboptimal supply then harms residential and industrial consumers across the income distribution. More broadly, the social norm that electricity is a right undercuts electricity access and reliability, holding back poverty reduction and ultimately economic growth.

The high-level policy recommendation is that electricity be treated as a private good. There are a variety of specific approaches to achieving this goal (Burgess et al. 2020), but they generally will not work unless there is a commitment to changing the social norm. Finally, it is important to emphasize that it is possible to protect the poor *and* treat electricity as a private good by, for example, providing direct transfers to low-income households. The design of politically feasible and efficient policies to protect low-income households from energy prices is an important area of research.

III. Conclusion

Climate change is having a moment and rightly so. But we would be much better off if the global energy challenge was instead having the moment—as it is perhaps the defining challenge of the century. Its essence is that, in choosing energy sources, we are choosing different combinations of the cost and reliability of energy, air quality, and climate risk. There is no one source of energy that dominates on all

[34] See Burgess et al. (2020) for a fuller description of the development of the social norm that electricity is a right and the resulting consequences.

[35] There are a variety of technologies (e.g., smart meters and prepayment smart meters) that aim to make electricity excludable so that payments and supply are tightly linked. Some of these efforts have succeeded, but many have failed precisely because they ultimately require breaking the powerful social norm that electricity is a right by cutting off those that do not pay.

Table 3—The Global Energy Challenge across Three Cities

	Oslo	Beijing	Lahore
Population (millions)	0.7	21.5	13.0
GDP per capita (2022 US$)	117,200	27,500	4,100
Annual per capita electricity consumption (kWh)	12,000	5,720	590
Life expectancy gain from reducing $PM_{2.5}$ to WHO standard (years)	0.2	3.2	> 6
End-of-century damages from 3° C warming (percent of end-of-century GDP)	−9%	2%	14%

Notes: Population data for Oslo come from Norway (2023), Beijing from National Bureau of Statistics of China (2021), and Lahore from Pakistan Bureau of Statistics (2023). GDP per capita for Oslo comes from OECD (2019) and is scaled to 2022 proportionally to national per capita GDP growth; Beijing comes from Beijing Municipal Bureau of Statistics (2010–2023); and Lahore is scaled from Pakistan's national GDP (World Bank 2024a) according to Lahore's contribution to national GDP estimated from Tahir (2017) assuming constant relative growth between Lahore and Pakistan. Annual per capita electricity consumption for Oslo is computed by dividing Oslo total retail electricity consumption by Oslo population, for Beijing by dividing Beijing electricity consumption by population, and for Lahore by dividing Punjab electricity consumption by Punjab population. Electricity consumption for Oslo is from Statistics Norway (2023) and Beijing and Punjab from CEIC (2023a, b). Gains in life expectancy from reducing $PM_{2.5}$ to the WHO standard are from Greenstone and Hasenkopf (2023). End-of-century damages correspond with 3.0° C of warming by end of century relative to preindustrial levels. End-of-century damages as a percent of end-of-century GDP is calculated as the sum of damages from 2080–2099 divided by the sum of GDP over that period. These estimates are obtained from ongoing and preliminary CIL analyses combining damages from agriculture, mortality, energy, coastal and sea level rise, and labor disutility across roughly 25,000 impact regions (Jina et al. 2024). Negative damages indicate benefits due to warming.

three of these dimensions, so the global energy challenge requires all societies and countries to make difficult trade-offs across these vital goals. Further, the trade-offs that govern these choices differ around the world based on wealth, geography, current air pollution concentrations, current and projected future climate, values, functioning of the political system, and so forth. What makes the global energy challenge even more complicated is that GHG emissions have the same impact on climate change wherever they are emitted—so in the end, each society's choices affect all other societies.

The nature of the global energy challenge's trade-offs and the interrelated consequences of each society's choices are illustrated by three important and very different cities. Table 3 reports that Oslo, Norway, has a population of 709,000 and per capita GDP of roughly $117,200. Its residents consume about 12,000 kWh of electricity per capita annually, air pollution is a small problem (reducing air pollution to the WHO standard would increase life expectancy by just 0.2 years), and 3.0° C warming at the end of the century is projected to produce climate *benefits* that are roughly equal to 9 percent of end-of-century GDP.[36] The global energy challenge is not very challenging in Oslo, as energy consumption and incomes are high, air quality is excellent, and the *local* effects of climate change are beneficial.[37] Thus, Oslo's likely goal, and probably the goal of many wealthy countries, is to help avoid disruptive climate change that might result from a high realized climate sensitivity without compromising too much on energy costs.

Now consider the 21.5 million people that live in Beijing, China, with its per capita GDP of roughly $27,500. Their current electricity consumption is 5,720 kWh annually, reducing air pollution to WHO standards would increase life expectancy by approximately 3.2 years, and 3.0° C end-of-century warming is projected to cause damages of 2 percent of end-of-century GDP. Their interests seem more likely to be centered on increasing energy consumption (and incomes) and reducing air pollution, while local climate concerns are unlikely to be substantial.

And then there is Lahore, Pakistan, home to 13 million people that have a per capita GDP of roughly $4,100. Today, annual per capita electricity consumption is just 590 kWh, air

[36] Population, per capita GDP, and electricity consumption sources vary by city and are documented in the notes of Table 3. Increases in life expectancy from PM reductions come from Greenstone and Hasenkopf (2023). Damages from warming are from ongoing work in Jina et al. (2024).

[37] Climate change's total impacts on Oslo are likely to differ from their direct local effects in several ways (e.g., the geopolitical consequences of global migration patterns), but these direct local impacts nevertheless likely influence local decision-making.

pollution is shortening life expectancies by more than 6 years, and end-of-century climate damages are projected at roughly 14 percent of end-of-century GDP. In Lahore, it undoubtedly seems critical to make progress on all three goals, but my observation is that increasing energy consumption and facilitating economic growth currently rises above the other ones.[38]

Thus, the inherent trade-offs in Oslo are very different from those in Beijing, and the trade-offs in Lahore don't resemble the ones in either of those cities. Yet the decisions that each of these places make about these trade-offs affect the others. The question then is whether there are paths where Oslo, Beijing, and Lahore all pursue their individual self-interest and where their choices do not make it unnecessarily difficult for each other, and perhaps even make it easier. This could take many forms, but two that seem promising are pricing particulates emissions in countries where air pollution is a great risk (which could have the global benefit of reducing GHG emissions by taxing fossil fuels) and wealthy countries funding technical and policy innovation that makes the global energy challenge's trade-offs less sharp,

More broadly, the best place to start in confronting the global energy challenge is to root out the many market failures that plague energy markets around the world. These include the failure to fully price externalities, a failure to adequately subsidize technical and policy innovation, information failures, and that energy has become a right in many places. However, history makes clear that it is not sufficient to name these market failures or the blackboard solutions that apply in frictionless worlds where winners compensate losers both within a country and across countries (even globally)—as this misses the nuances and politics of the real world.

Effective solutions to energy market failures lie in the plumbing that uses economics to diagnose the problems and find successful policy reforms. On the former, this means quantifying trade-offs and learning about the grimy mix of market structures, incentives of key stakeholders and actors, institutional structures, opportunities for redistribution, and the extent of state capacity that policy must run through. On the latter, policy to correct market failures is much more likely to be successful if it is paired with relentless testing and experimentation.

What role is there for economists and the economics profession? We can do more than declare the mission accomplished after shouting "price externalities," "subsidize public goods," and "do not let private goods become rights." Among our potential contributions, I would put identifying politically feasible solutions, testing their implementation, and helping governments scale them up at the top of the list. The payoff is that we can help to make the global energy challenge less challenging.

[38] Oslo, Beijing, and Lahore all have high incomes relative to the rest of their countries. For example, mean 2022 GDP per capita is $106,200 in Norway, $12,700 in China, and $1,600 in Pakistan (World Bank 2024b).

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